

Assignment 9

(Assume all rings are commutative with multiplicative identity)

1. Show that any ring automorphism of $\mathbb{R}$ is identity.
2. If R is an integral domain, compute the unit group of $R[X]$ and $R[[X]]$.
3. Let $f : R \rightarrow R'$ be a ring homomorphism. Show that if I is an ideal in R' , then $f^{-1}(I)$ is an ideal in R . Also show that if I is a prime ideal, so is $f^{-1}(I)$. Is the same true for maximal ideals ?
4. Let X be a metric space and let $p \in X$ be a point. Let $\mathcal{C}(X)$ be the ring of all continuous real-valued continuous functions on X . Show that :
 - a) $\mathfrak{m}_p = \{f \in \mathcal{C}(X) \mid f(p) = 0\}$ defines a maximal ideal in $\mathcal{C}(X)$.
 - b) Show that if X is compact, every maximal ideal is of this type.
5. Give an example of a ring which is not a field but has a unique maximal ideal.